# Big Data to Information (AIT 580-001)
# Data Analytics Research Paper

## Predicting and Reducing Greenhouse Gases emitted by the Petroleum Industry Using Machine Learning and Data Analytics.

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***Abstract***
*This project explores and analyses the greenhouse gas emissions using two datasets: "Supply Chain Greenhouse Gas Emission Factors" and "Greenhouse Gas Reporting Program 2023"dataset. The aim of this study is to analyse and examine the emission patterns of the greenhouse gases of different facilities, the role of facility attributes, and petroleum-related industry's carbon intensity per $1. This study leverages multiple tools such as Python, R, SQL, AWS S3, AWS EC2, AWS RDS and AWS Glue DataBrew to perform cleaning, exploration, analysis, and visualization of the data.*

***Keywords: Greenhouse gas emissions, analyze, explore, petroleum-related industry, carbon intensity, cleaning, Python, R, SQL, AWS***

## 1. Introduction

The earth is located in a favourable position in the solar system, it receives just enough sunlight from the sun and its atmosphere is capable of trapping this energy to maintain surface temperature to sustain life. The process in which the heat gets trapped by the atmosphere is called the "greenhouse effect" and the gases being trapped are known as "greenhouse gases" [1]. The gases consist of carbon dioxide, methane, ozone, nitrous oxide, chlorofluorocarbon [1]. CO2 plays a vital role in the sustenance of life on Earth. However, in recent years, there has been an increase in the greenhouse gases levels mainly due to human activities [1]. Activities such as burning of fossil fuels (petroleum industry) and deforestation have increased the concentration of CO2 [2]. Methane is a mix of natural and human-driven sources. It is produced when the plant materials get broken down. Methane leaks during fossil fuel extraction and processing is a huge factor for the increase in this emission [2]. Nitrous Oxide is also an extremely impactful gas and is mainly released in agricultural activities. It is also produced when the fossil fuels are burnt [2]. Its concentration in the atmosphere has risen by around 18 in the last century [2].

## 2. Research Questions

1) Is the facility data available in the current dataset sufficient for an Artificial Intelligence driven prediction to reduce direct emissions of greenhouse gases while still maintaining energy efficiency?

2) Do facility attributes have an impact on the emission of greenhouse gases? If so, how does their effect vary?

3) How much greenhouse gas is emitted per $1 in the petroleum industry? And how does it compare to other industries emitting greenhouse gases per $1?

4) Are there any identifiable patterns among facilities based on their greenhouse gas emissions?

## 3. *Literature Review*

### 3.1 A Review of Carbon Accounting in Petroleum Refining Industry [3].

In this paper, the authors discuss how the carbon emissions in the petroleum refining industry is calculated and why it is essential to account for this to reduce global carbon emissions for the betterment of the environment. Xing et al. (2025) review the existing methods to calculate the carbon emissions. Some of the mentioned methods are emission factor method (multiplying fuel consumption by fixed coefficients), AI-based methods (also known as data-driven), mass balance and more [3].

The report discusses how most petroleum refineries primarily rely on the static emission factors which can be the leading cause for errors in carbon emission data tracking and how this issue can be resolved by creating a framework which uses the continuous data from the units in the refinery to calculate the emissions being emitted in near-real-time. The framework will be developed using artificial intelligence and automation to process and analyze the data from the units of the refinery which will help provide accuracy and precision. The authors also emphasize how machine learning and data-driven models can help refineries and industries understand which particular units produce the most amount of carbon emissions so that proper measures can be taken in the future to reduce these emissions[3].

I chose this research report because it confirms my work regarding using artificial intelligence to predict and reduce the greenhouse gases emitted in the petroleum industry. This paper also relates well with one of my research questions i.e, “How can Artificial Intelligence driven prediction reduce direct emissions of greenhouse gases while still maintaining energy efficiency?” as it discusses how artificial intelligence and data-driven models and techniques can process large amounts of data produced by each individual unit and analyze it to predict carbon emissions precisely and also optimize energy use.

### 3.2) Uncovering the Lifecycle Carbon Emissions and Its Reduction Pathways: A Case Study of Petroleum Refining Enterprise [4].

In this research paper, the authors examine the lifecycle of the carbon emissions being emitted by petroleum refineries and discuss different pathways to reduce these emissions [4]. Zhao et al. (2024) discuss using two models (1) Life Cycle Assessment and (2) System Dynamics to measure the quantity of the emissions being produced and then predict future trends based on the results. Since petroleum refineries have multiple processes and activities, the authors divide the emissions into 5 different categories [4]. These categories are:

1) Fuel Combustion
2) Industrial process emission
3) Flaring
4) Purchased Steam
5) Fugitive Emissions

The authors also found out that fuel combustion and some industrial processes produce the most amounts of carbon emissions. With the help of the model, they run different scenarios which help determine/predict the achievement of carbon neutrality. These technical scenarios are based on use of clean-energy, carbon-capture technology and improvements of the machines for efficiency in the energy. Their model predicts (based on the ideal scenarios) carbon neutrality could be achieved around 2058 in China [4].

This research report focuses deeply on the prediction and reduction of the emissions in the petroleum industry while maintaining efficiency and is aligned with my research. My second research question "Do facility attributes have an impact on the emission of greenhouse gases? If so, how does their effect vary?" is answered in this report as the authors demonstrate the impact refinery layout plans and organisation plays for the emissions. Their models display results on how changes in the operational strategy plays an important role in the emission of carbon from different processes.

### 3.3) Structural Evolution of China's Intersectoral Embodied Carbon Emission Flow Network[5]

In this final research paper, the authors discuss how the carbon emissions emitted by the petroleum industry are transferred to other industries in China. That is, they discuss how all the industrial sectors are somehow connected in a way [5]. Wang et al. (2021)studied the input-output table of China's economy in the years 2002, 2005, 2007, 2010, 2012 and 2015 and covered around 30 different industries. This input-output table contained information about how much was being bought/sold by each industry to each other. This data was later utilized to build 6 "indirect" carbon emission flow networks (1 for each year). With the use of complex network analysis, they were able to find how each industry was connected to each other and hence had an influence on each other. Coal and petroleum industries/sectors were the major supply of "indirect" carbon emission and industries like construction, manufacturing and service were their main "consumers". Therefore improvements in major

supplier industries like coal, petroleum, fuel processing and more could positively impact the emission of carbon as they influence the other industries [5].

This paper provides information regarding the interconnection of different industries and how the petroleum industry plays a huge role in the economy and it has an influence on the other industries with high emission of carbon. Although it doesn't align completely with my third research question "How much greenhouse gas is emitted per $1 in the petroleum industry? And how does it compare to other industries emitting greenhouse gases per $1." This study helps in understanding why the petroleum industry should be focused on. We can also understand which industry (other than the petroleum) emits more carbon emissions as compared to their economic activity [5]. By using the NAICS data, I can understand and identify key emission sectors and compare their greenhouse gases emissions to the petroleum industries emissions and understand the interconnection of the industries in America.

## *4. Methodology*

This study leverages a multi-tool analytical and exploratory approach by utilizing Python, R, SQL, AWS EC2,S3 and AWS Glue DataBrew. These tools are employed to clean, explore, transform and analyze the datasets. This section outlines how each tool was utilized and how the data will be used to answer the research questions.

### 4.1) Dataset Description

a) Supply Chain Greenhouse Gas Emission Factors v1.3 by NAICS-6 (2022):

This dataset provides details on how much greenhouse gas is emitted for $1 activity [6]. It has 1017 rows and 9 columns.

b) Greenhouse Reporting Program 2023 data:

This dataset provides detailed information about US facilities which emit greenhouse gases, emission from processes. This can help in making better analysis and discovery of high emission areas [7]. This dataset has 7 subsections.

Number of records and columns:

- Direct Point Emitters has 6470 rows and 17 columns
- Onshore Oil and Gas Production has 445 rows and 17 columns
- Gathering and Boosting has 349 rows and 17 columns
- Transmission Pipeline has 164 rows and 16 columns
- LDC- Direct emissions has 157 rows and 17 columns
- SF6 from Electrical Emissions has 81 rows and 14 columns
- Suppliers has 1004 rows and 19 columns

In this study, the "Direct Point Emitters" subsection will be used for a detailed statistical analysis.

### 4.2) Data Exploration using Python

For the data exploration in python, after uploading the "Greenhouse Reporting Program 2023 data" dataset and cleaning the dataset to ensure there were no missing or invalid values, I explored various visualizations regarding the distribution of the emissions, number of facilities in each state, dispersion of total emissions across different latitudes and a correlation

heatmap of the different emission types, we are able to identify important trends and patterns which would help answer our research question.

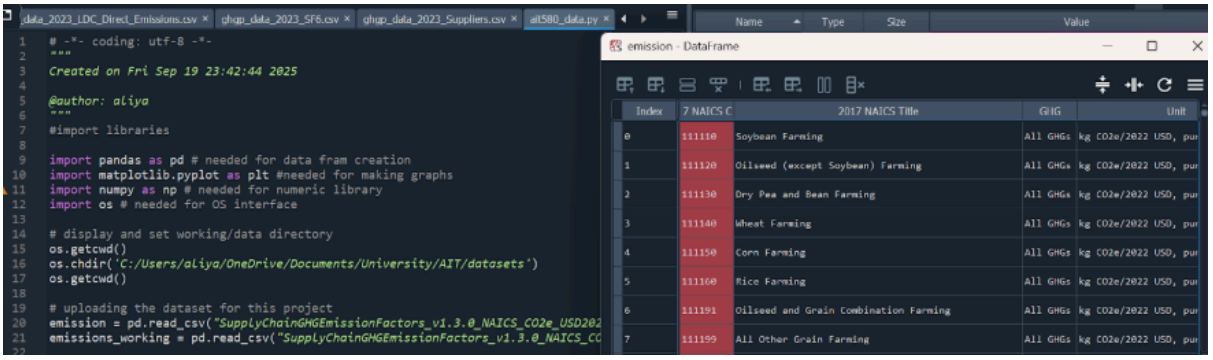


Fig.1 Uploading dataset in spyder for exploration

### 4.3) Data Exploration using R

We used R for additional data analysis and exploration. We utilized a random forest regression model which was initially trained and then used to test if the facility variables/attributes available in the current dataset would be sufficient to predict emissions using AI driven tools. This helps us answer our first research question. Variable importance measures were also examined to assess the significant relevance of the predictors (State, NAICS code, Industry type and more). We also leveraged the K-means clustering model to identify the facilities emission profiles. This helps us answer our fourth research question.

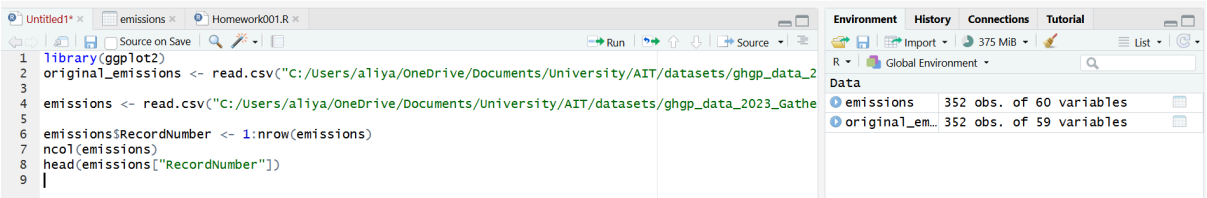


Fig. 2 Uploading dataset in RStudio for exploration

### 4.4) Data Exploration using SQL

I utilized the supply chain greenhouse gas dataset for data exploration in SQL. I first imported the dataset into a relational database using MariaDB and created a table and ran multiple SQL queries to explore and analyze the number of facilities in the dataset and understood the NAICS summaries.

ghg.supplychain_factors: 2,012 rows total (approximately), limited to 1,000 — Next — Show all — Sorting — Columns (8/8) — Filter

| # | 1 naics_2017_code | 2 naics_2017_title | 3 ghg | 4 unit | 5 ef_without_margins | 6 ef_margins | 7 ef_with_margins | 8 useeio_ref_cod |
|---|---|---|---|---|---|---|---|---|
| 1 | 111,110 | Soybean Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.488 | 0.044 | 0.532 | 1111A0 |
| 2 | 111,120 | Oilseed (except Soybean) Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.488 | 0.044 | 0.532 | 1111A0 |
| 3 | 111,130 | Dry Pea and Bean Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.809 | 0.04 | 0.848 | 1111B0 |
| 4 | 111,140 | Wheat Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.809 | 0.04 | 0.848 | 1111B0 |
| 5 | 111,150 | Corn Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.809 | 0.04 | 0.848 | 1111B0 |
| 6 | 111,160 | Rice Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.809 | 0.04 | 0.848 | 1111B0 |
| 7 | 111,191 | Oilseed and Grain Combination Farmi... | All GHGs | kg CO2e/2022 USD, purchaser price | 0.809 | 0.04 | 0.848 | 1111B0 |
| 8 | 111,199 | All Other Grain Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.809 | 0.04 | 0.848 | 1111B0 |
| 9 | 111,211 | Potato Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.591 | 0.041 | 0.631 | 111200 |
| 10 | 111,219 | Other Vegetable (except Potato) and ... | All GHGs | kg CO2e/2022 USD, purchaser price | 0.591 | 0.041 | 0.631 | 111200 |
| 11 | 111,310 | Orange Groves | All GHGs | kg CO2e/2022 USD, purchaser price | 0.429 | 0.059 | 0.488 | 111300 |
| 12 | 111,320 | Citrus (except Orange) Groves | All GHGs | kg CO2e/2022 USD, purchaser price | 0.429 | 0.059 | 0.488 | 111300 |
| 13 | 111,331 | Apple Orchards | All GHGs | kg CO2e/2022 USD, purchaser price | 0.429 | 0.059 | 0.488 | 111300 |
| 14 | 111,332 | Grape Vineyards | All GHGs | kg CO2e/2022 USD, purchaser price | 0.429 | 0.059 | 0.488 | 111300 |
| 15 | 111,333 | Strawberry Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.429 | 0.059 | 0.488 | 111300 |
| 16 | 111,334 | Berry (except Strawberry) Farming | All GHGs | kg CO2e/2022 USD, purchaser price | 0.429 | 0.059 | 0.488 | 111300 |

Fig. 3 Uploading dataset in MariaDB for exploration

### 4.5) Data Exploration using AWS EC2/ S3 / RDS

I used Amazon Web Services such as EC2 (Amazon Elastic Compute Cloud) and a S3 (Amazon Simple Storage Service )bucket to store and host my dataset in the cloud and run python scripts for analysis. Later, I used AWS RDS (Amazon Relational Database Service) to host my SQL database in the cloud which was essential in enabling a secure connection and scalability.

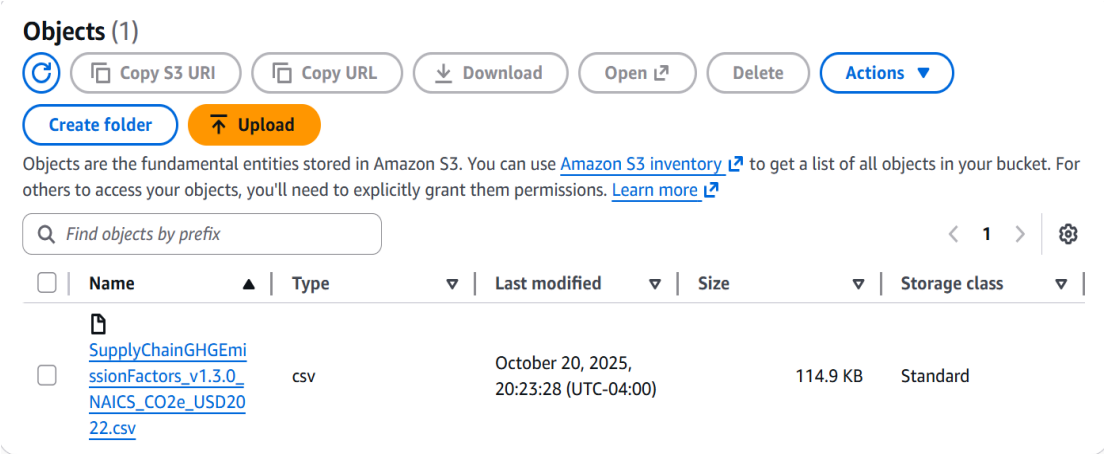


Fig. 4 Uploading dataset in S3 bucket

### 4.6) Data Exploration using AWS Glue DataBrew

I used the Amazon Glue DataBrew for direct visualization and analysis. It automatically helps in detecting missing/invalid values, outliers and generates summary statistics which aids in analysis.

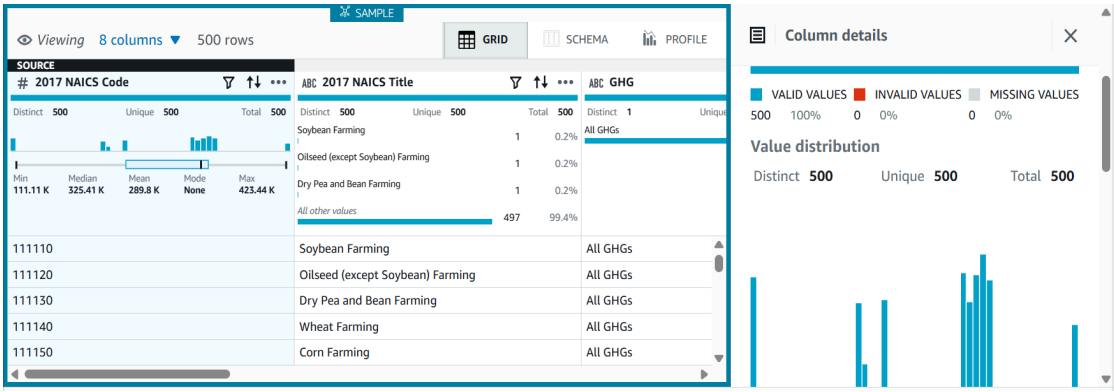


Fig. 5 Uploading and analyzing project in AWS Glue DataBrew

### 4.7) NOIR datatypes

For the data exploration and analysis, I also examined the four NOIR datatypes (Nominal, Ordinal, Interval and Ratio) using the variables from the dataset.

a) Nominal: Primary NAICS code.

The North American Industry Classification Code is a 6 digit code which helps in the identification of different businesses and industries [NAICS]. This helps in understanding the number and type of industries in the dataset.

b) Ordinal: Intensity of Supply Chain Emissions.

This variable was created based on the "Supply Chain Emission Factors with Margins" from the original dataset. This variable helps in identifying the facilities with the highest emission levels (Low, Medium, High, Very High).

c) Interval: Year

This variable was also created as there were no interval datatypes available in the dataset. This variable indicates the year during which the records were recorded.

d) Ratio: Total_Emissions

This variable helped in analyzing and visualising the quantity of emissions. Since the data had a true zero, this variable had meaningful comparison.

## *5. Data Analysis and Results*

In this section, I will discuss the analysis conducted for each research question. I will also describe each approach applied to obtain the results.

Q1) Is the facility data available in the current dataset sufficient for an Artificial Intelligence driven prediction to reduce direct emissions of greenhouse gases while still maintaining energy efficiency?

To evaluate if the available facility data is sufficient for an AI driven prediction to reduce direct emissions of greenhouse gases, I utilized a Random Forest regression model. This model was developed using the greenhouse reporting program 2023 dataset. This model would help in predicting the total greenhouse gas emitted directly from each facility using predictors such as state, latitude, longitude, the NAICS code, the industry subpart classification and total emissions.

```
emission2<- emission[, c("State.where.Emissions.Occur",
              "Reported.Latitude",
              "Reported.Longitude",
              "Primary.NAICS.Code",
              "Industry.Type..subparts.",
              "Total.reported.direct.emissions.from.Local.Distribution.Companies")]

emission2<- na.omit(emission2)
names(emission2) <- c("State", "Latitude", "Longitude",
                "NAICS", "Industry", "Total_Emissions")
```

Fig. 6 Uploading the dataset

After running the model, I found:

```
library(randomForest)
set.seed(1)

fit1 <- randomForest(Total_Emissions ~ State + Latitude + Longitude +
                       NAICS + Industry,
                     data = emission2,
                     ntree = 150,
                     importance = TRUE)
print(fit1)
importance(fit1)
varImpPlot(fit1)
```

Fig. 7 Running the random forest model

- % Variance explained:  -0.44
- Mean of squared residuals: 17173829698

```
> importance(fit1)
              %IncMSE IncNodePurity
State      -1.52450507  157188376715
Latitude   -0.12691981  253457728421
Longitude  -0.43845193  471391472405
NAICS       0.08342887    8696947806
Industry    0.00000000    6562586203
```

Fig. 8 Variable importance

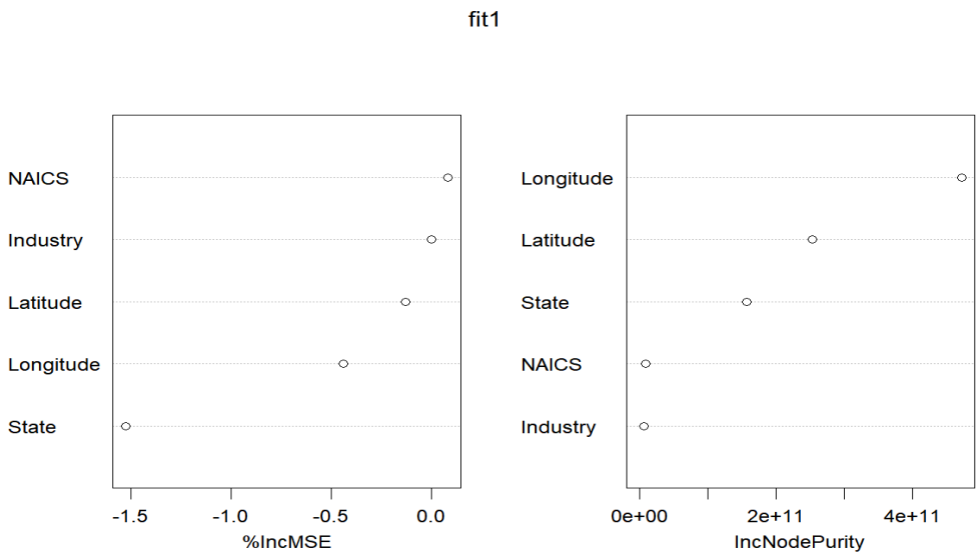


Fig. 9 Fit 1

In the left graph (%IncMSE), NAICS seems to be the only positive variable. The rest are negative and it indicates the model performed worse. Therefore, the variables/ attributes in the current dataset are not strong predictors for estimating the total emissions. The current attributes are not enough for AI driven predictions and may need other variables such as real-time operational data.

This analysis demonstrates that the current attributes are insufficient for AI driven emission prediction as real-time data processing and continuous monitoring are required. This finding is consistent with Xing et al. (), who highlighted that AI based emission prediction requires continuous monitored data to achieve accuracy.

Q2) Do facility attributes have an impact on the emission of greenhouse gases? If so, how does their effect vary?

To answer this question, I utilized the greenhouse reporting program dataset to analyze and understand if the attributes or a facility has an impact on the emission of the greenhouse gases. Both univariate and multivariate analysis were conducted to assess this.
Initially, I identified the columns from the Direct Emitters subsection from the dataset.
Independent variables:
State, Latitude, Longitude, Industry, CO2, CH4,N2O
Dependent variable: Total_emission

```
emission = pd.read_csv("ghgp_data_2023_LDC_Direct_Emissions.csv", skiprows=3)
emission.columns = emission.columns.str.strip()


emission = emission.rename(columns={
    'State where Emissions Occur': 'State',
    'Reported Latitude': 'Latitude',
    'Reported Longitude': 'Longitude',
    'Industry Type (subparts)': 'Industry',
    'Total reported direct emissions from Local Distribution Companies': 'Total_Emiss
    'CO2 emissions (non-biogenic)': 'CO2',
    'Methane (CH4) emissions': 'CH4',
    'Nitrous Oxide (N2O) emissions': 'N2O'
})


emission = emission.drop(columns=[col for col in emission.columns if "Unnamed" in col


emission = emission.dropna(subset=['Facility Id'])


for col in ['Total_Emissions', 'CO2', 'CH4', 'N2O', 'Latitude', 'Longitude']:
    emission[col] = pd.to_numeric(emission[col], errors='coerce')
```

Fig. 10 Uploading the dataset and cleaning it for analysis

*Univariate analysis*

**Distribution of Total Direct Emissions**

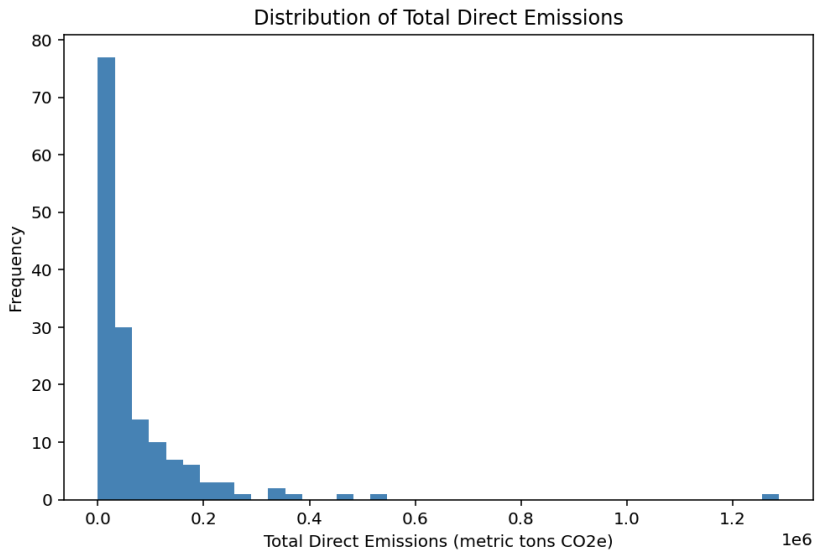


Fig. 11 Distribution of direct emissions histogram

As we can see, this histogram shows that the total direct emissions are highly right-skewed. That is, most of the facilities in the US emit low amounts of greenhouse gases (majority under 200,000 metric tons CO2e), whereas a small number of facilities have a higher emission and with one facility exceeding 1.2 million metric tons.

This skewed distribution helps in understanding that the emissions are primarily produced by a small group of large industrial facilities which implies considerable variability across the dataset.

**Facilities by state**

The following bar chart clearly shows the number of facilities in each state. States such as Tennessee, New York have 8+ facilities and states like Georgia, Utah, Oregon, Vermont have less than 2. Therefore, state-level variation may also play a role in the emission distribution as some states may have better topological attributes that influence the number of facilities.

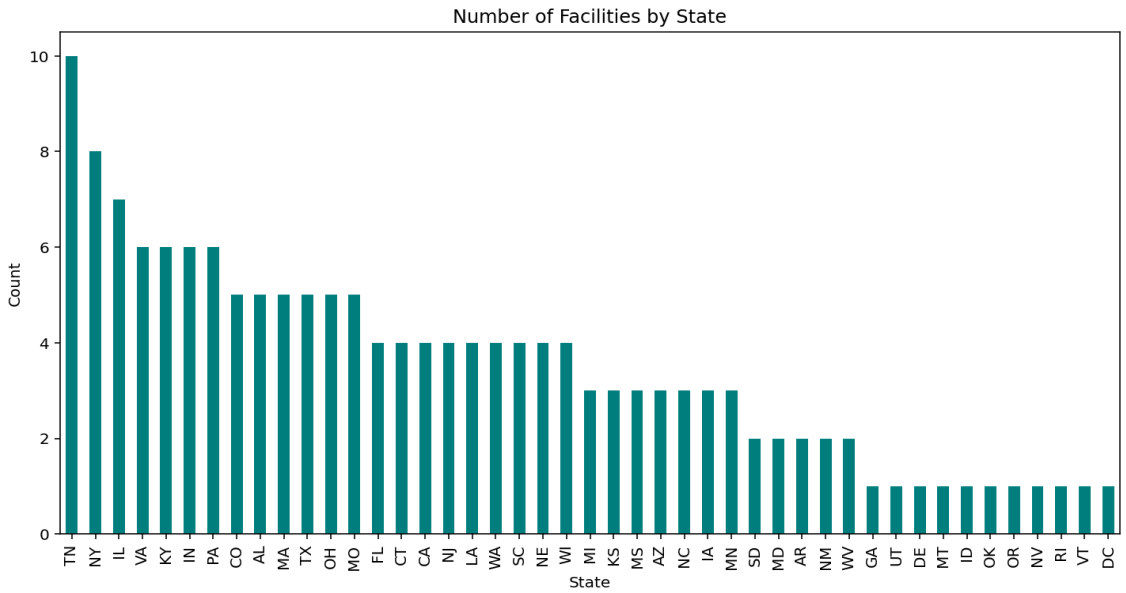


Fig. 12 Barchart of number of facilities in a state

*Multivariate Analysis*

**Total emissions vs Latitude**

The following scatterplot indicates a dispersion of total emissions across different latitudes.

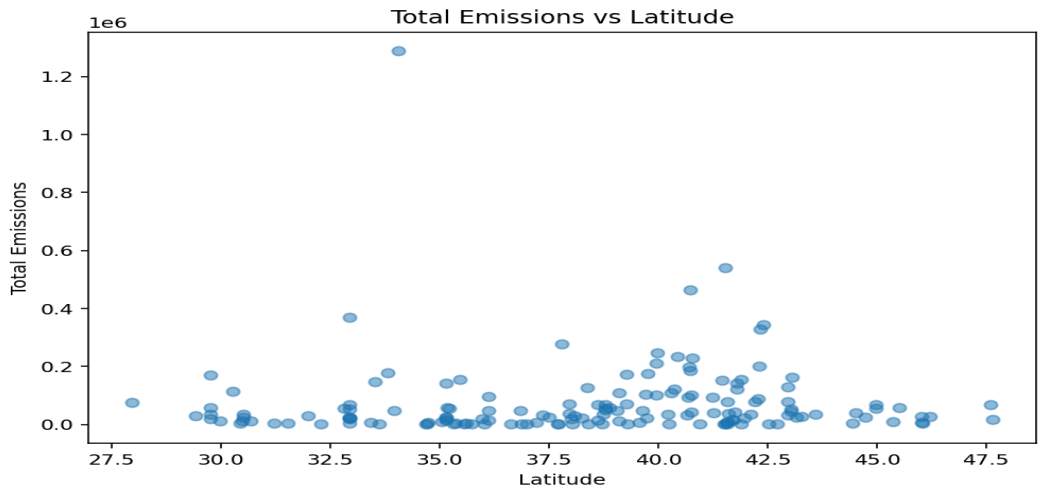


Fig. 13 Scatterplot of Total Emissions vs Latitude

We can see that there is a lot of concentration between 30.0 and 45.0 latitude. Very few lie below 30.0 and a few lie above 45.0. From [9], I was able to understand almost all the states lie between 30.0 to 45.0 latitude. Although emissions can not be determined by latitude alone, regional industrial activity may play a role in the emissions.

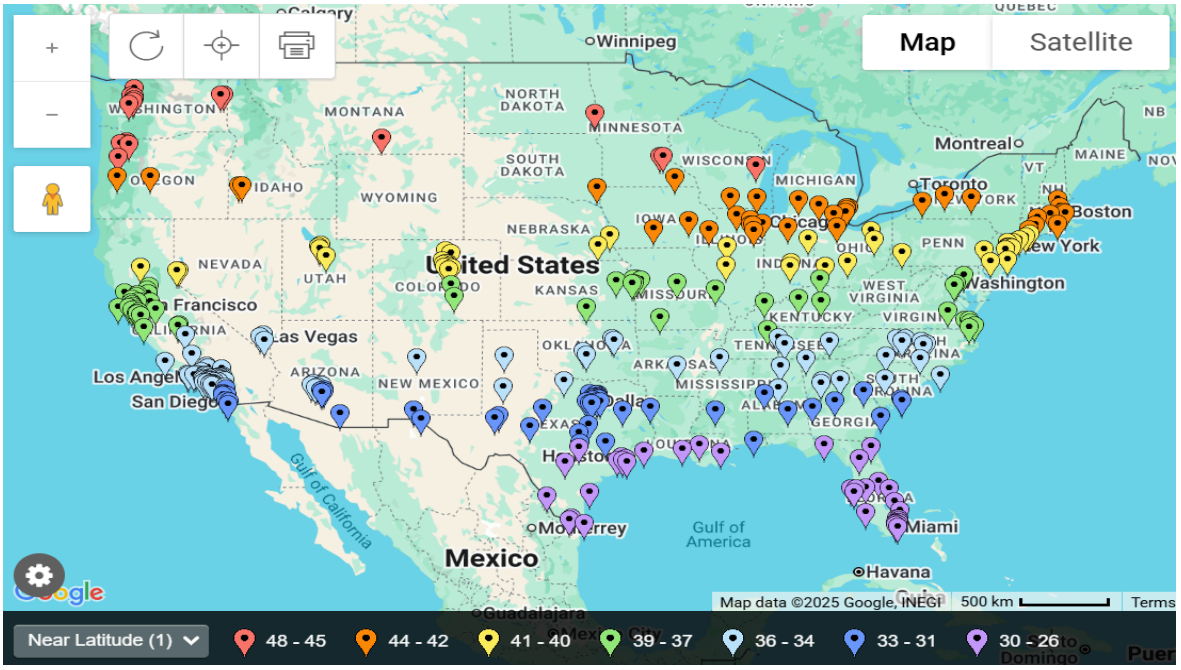


Fig. 14 Latitude of each state.

**Correlation Heatmap of Emission Types**

The following heatmap is essential as it reveals important insights.

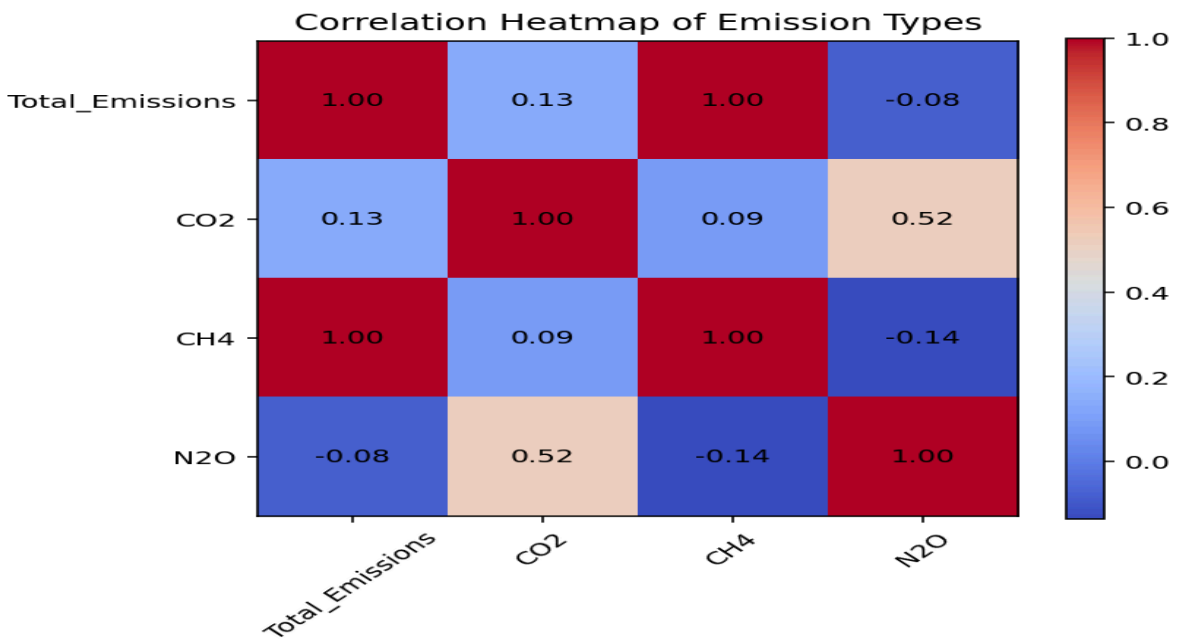


Fig. 15 Correlation Heatmap

- The Total emission and CH4 indicate a perfect correlation of 1.00

This means that for all the industries, methane emissions are the primary contributors.

- CO2 and N20 are moderately correlated as their correlation level is 0.52

This means facilities which emit more CO2 might also be emitting more N20.

- Total emission does not have a strong correlation with both CO2 (0.13) and N2O (-0.08).

This means that the total emission is mostly affected by the emission of methane rather than carbon dioxide and nitrogen oxide.

This helps us understand the type of emission varies in each facility and facility attributes like state and latitude might have an indirect influence on the emission of gases. The industry type and emissions composition provide more insight as some may emit high amounts of CH4 or CO2 and N2O. The main attributes for understanding emission levels are industrial processes.

Q3)How much greenhouse gas is emitted per $1 in the petroleum industry? And how does it compare to other industries emitting greenhouse gases per $1.

For this research question, we will use the supply chain dataset to measure the carbon intensity across multiple petroleum related industries and compare them. We first select industries with certain keywords to ensure they are from or related to the petroleum industry. We choose keywords such as "petroleum, crude, natural gas, pipeline, refining, drilling and gasoline"

```
emission = pd.read_csv("SupplyChainGHGEmissionFactors_v1.3.0_NAICS_CO2e_USD2022.csv")

emission.columns = emission.columns.str.strip()


petroleum_emission = emission[emission['2017 NAICS Title'].str.contains("petroleum|crude|natural gas|r

petroleum_sorted = petroleum_emission.sort_values("Supply Chain Emission Factors without Margins", asc

petroleum_sorted[['2017 NAICS Title',
                  'Supply Chain Emission Factors without Margins']].head(15)
```

Fig. 16 Selection of keywords for visualization

**Petroleum industry emissions per dollar**

The following graph helps us identify each petroleum-related industry and its greenhouse gas emission per $1. We are able to understand that several sectors have a high carbon intensity especially, pipeline transportation which exceeds 1 kg CO2e per $1. Other sectors such as crude petroleum extraction and natural gas extraction are also high carbon intensive. Comparatively, petroleum refineries and gasoline stations have a lower carbon intensity.

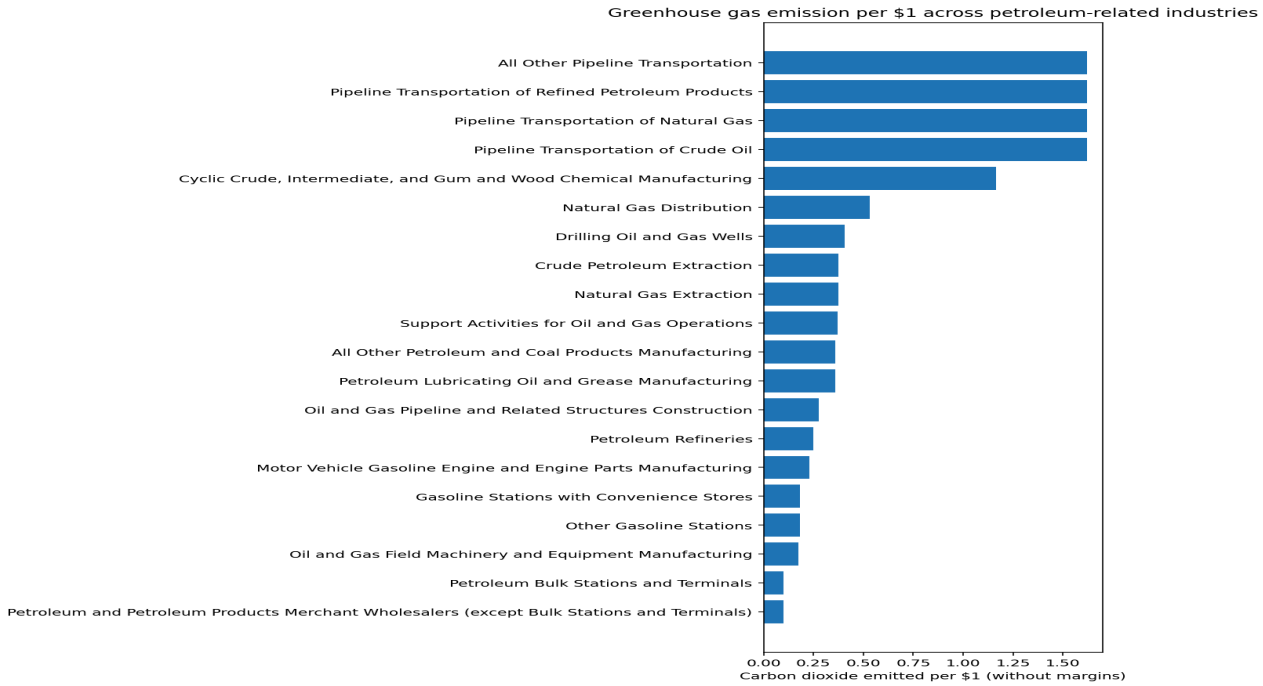


Fig. 17 Greenhouse gas emissions per $1

**Comparison to other industries**

The barchart below shows the difference in the average CO2e per $1 between petroleum related industries and other industries. As we can see, petroleum industries produce an average of 0.6 kg C02e per $1 whereas the other industries produce around 0.3 kg C02e per $1 which shows that the petroleum industry is twice as carbon intensive.

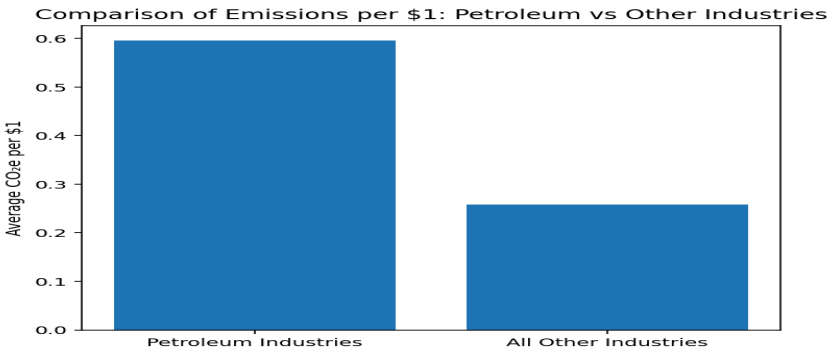


Fig. 18 Comparison of emissions

Q4) Are there identifiable clusters or patterns among facilities based on their emission profiles ($CO_2$, $CH_4$, $N_2O$, and Total Emissions)?

In this research question, we will try to explore any identifiable patterns in the emission behaviour of different facilities. By analyzing the main greenhouse gases, we can do cluster analysis to find patterns.

We will use the greenhouse reporting program dataset to answer this question.

```r
cluster_emdata <- emission[, c(
  "Total.reported.direct.emissions.from.Local.Distribution.Companies",
  "CO2.emissions..non.biogenic.",
  "Methane..CH4..emissions",
  "Nitrous.Oxide..N2O..emissions"
)]

names(cluster_emdata) <- c("Total_Emissions", "CO2", "CH4", "N2O")

# Remove rows with missing values
cluster_emdata <- na.omit(cluster_emdata)
cluster_scaled <- scale(cluster_emdata)
```

Fig. 19 Code for creating cluster

I utilized a K-means clustering algorithm to find any hidden patterns or groups. I extracted the greenhouse gases and a Total emissions variable.

```r
set.seed(2)
k3 <- kmeans(cluster_scaled, centers = 3, nstart = 20)

k3$centers

plot(cluster_scaled[, "CO2"], cluster_scaled[, "CH4"],
     col = k3$cluster, pch = 16,
     xlab = "CO2 (scaled)",
     ylab = "CH4 (scaled)",
     main = "K-Means Clustering of Facilities by Emission Profile")
```

Fig. 20 Code for K-means model

```
> k3$centers
  Total_Emissions        CO2        CH4        N2O
1      -0.2945461 -0.2553833 -0.2672009 -0.3645343
2       2.4250352 -0.3673000  2.4621184 -0.4317606
3      -0.1987727  1.6800833 -0.3769841  2.3287594
```

Fig. 21 Result of K3$centers

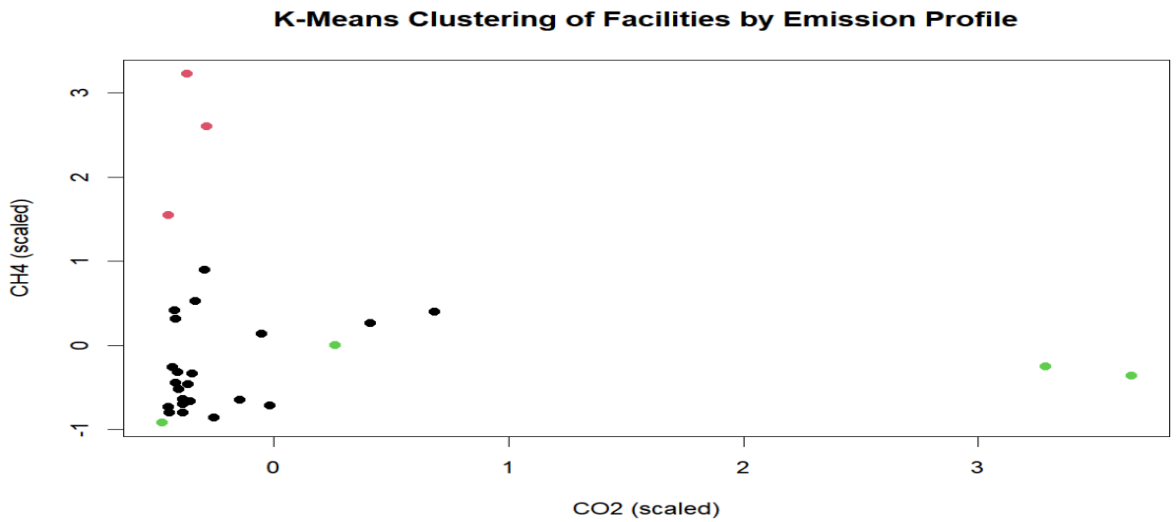


Fig. 22 K-means clustering of facilities.

Cluster 1: Low emission facilities (Black points)

- All emission variables are below average.
- Represents the majority of facilities.

Cluster 2: High methane facilities (Pink points)

- Extremely high methane emissions.
- Total emissions are also very high.
- Carbon dioxide and nitrogen oxide remain below average.

Cluster 3: High carbon dioxide and high nitrogen oxide facilities (Green points)

- Carbon dioxide emissions are high.
- Nitrogen oxide emissions are high.
- Methane is below average.

We can see that the emissions are not just randomly distributed but the facilities have three different emission behaviours.

## *6. Results*

By going through each research question and analyzing the data, I was able to understand that this dataset alone is not sufficient for AI-based predictions as there are not enough attributes required (such as, continuous real-time data, operational level data). This study also shows that while some facility attributes may indirectly affect the emission levels, their impact may vary. Industry type and emissions composition plays a bigger role than the latitude or longitude of a facility. We also find out that the petroleum industry emits more than double the greenhouse gases per $1 as compared to other industries and lastly, we are able to analyze that emission behaviour of an industry is not random but depends on industrial differences.

## *7. Limitations and Future works*

### 7.1) Limitations:

Although this study provides meaningful insight regarding the greenhouse gas emissions, it also has quite a few limitations.

The current dataset lacks important variables such as operational data and real-time production data which is important for AI-based predictions. Some variables such as the physical attributes of a facility may not play a strong role in truly understanding the emissions.

### 7.2) Future works:

Future analysis would benefit by utilizing operational data and continuously monitored data as those variables would help in predicting emissions using AI-driven models.

Increasing the scope of the data and including all the subsectors may all be beneficial as it increases the size of data and makes the data richer. Additionally, multi-year emissions would help identify trends

## *8. Conclusion*

This study explored greenhouse gas emissions using 2 datasets and utilized multiple tools such as Python, R, SQL, AWS, and machine learning techniques to answer four research questions. The results show:

- The current facility-level dataset is not sufficient for AI-based emission prediction, as key predictors required for accurate modeling are missing.
- Facility attributes may indirectly affect emissions, but their influence varies by the type of gas as methane plays the largest role in overall emissions.
- Petroleum-related industries emit more greenhouse gases per dollar than other sectors.
- K-means clustering model revealed 3 distinct emissions profiles.

Overall, the findings of this study focuses on the need for more detailed, real-time data to support AI-driven emission predictions. The results also highlight the potential for specific measures in high methane and petroleum intensive industrial areas.

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